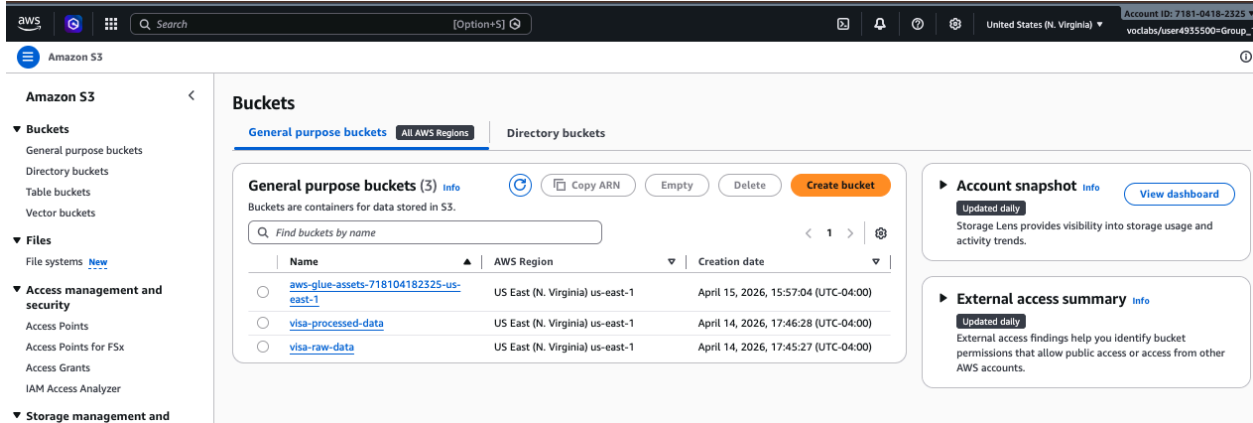


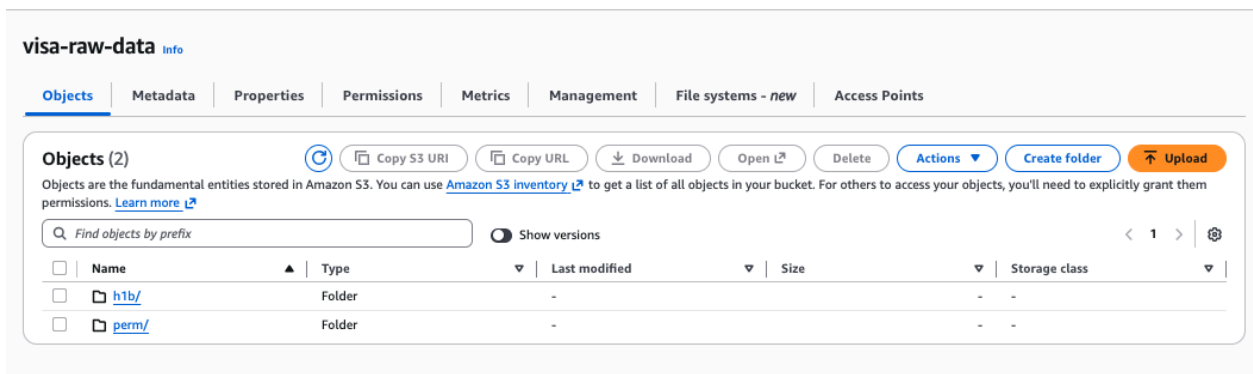
Amazon S3 Bucket Overview

Three S3 buckets used in the project are listed: visa-raw-data (stores raw input data), visa-processed-data (stores transformed data), and aws-glue-assets (auto-generated for Glue job artifacts), all located in the us-east-1 region.



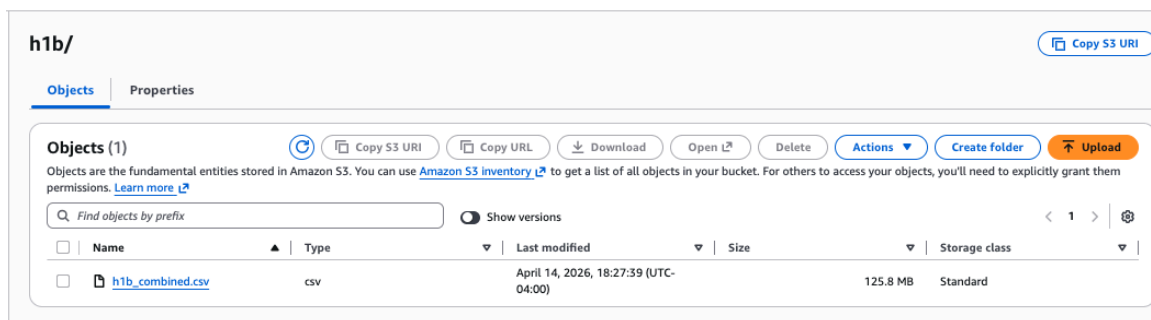
S3 Bucket – visa-raw-data Contents

The visa-raw-data bucket contains two folders: h1b/ and perm/, each storing raw visa application data by category, which serve as the source input for downstream Glue ETL jobs.



S3 Raw Data – h1b/ Folder Contents

The h1b/ folder contains a single CSV file h1b_combined.csv (Standard storage class), which serves as the raw H-1B visa application dataset uploaded for processing.



S3 Raw Data – perm/ Folder Contents

The perm/ folder contains perm_combined.csv (Standard storage class), which serves as the raw PERM labor certification dataset uploaded as the pipeline's second data source.

The screenshot shows the Amazon S3 console interface for the bucket 'visa-raw-data' and the folder 'perm/'. The left sidebar shows the navigation menu with 'Buckets' selected. The main content area shows the 'perm/' folder details. The 'Objects (1)' section displays a table with one object: 'perm_combined.csv'.

Name	Type	Last modified	Size	Storage class
perm_combined.csv	csv	April 14, 2026, 21:02:11 (UTC-04:00)	655.0 MB	Standard

S3 Bucket – visa-processed-data Contents

The visa-processed-data bucket contains two folders, h1b_clean/ and perm_clean/, which store the cleaned and transformed outputs written by the Glue ETL jobs after processing the raw data.

The screenshot shows the Amazon S3 console interface for the bucket 'visa-processed-data'. The left sidebar shows the navigation menu with 'Buckets' selected. The main content area shows the 'visa-processed-data' bucket details. The 'Objects (2)' section displays a table with two folders: 'h1b_clean/' and 'perm_clean/'.

Name	Type	Last modified	Size	Storage class
h1b_clean/	Folder	-	-	-
perm_clean/	Folder	-	-	-

S3 Processed Data – h1b_clean/ Folder Contents

The h1b_clean/ folder contains two Parquet files output by the Glue ETL job, stored in Snappy-compressed format, indicating that the cleaned H-1B data was partitioned across multiple files for efficient querying.

The screenshot shows the Amazon S3 console interface for the bucket 'visa-processed-data' and the folder 'h1b_clean/'. The left sidebar shows the navigation menu with 'Buckets' selected. The main content area shows the 'h1b_clean/' folder details. The 'Objects (2)' section displays a table with two Parquet files.

Name	Type	Last modified	Size	Storage class
run-AmazonS3_node1776281795861-1-part-block-0-r-00000-snappy.parquet	parquet	April 26, 2026, 20:47:53 (UTC-04:00)	11.3 MB	Standard
run-AmazonS3_node1776281795861-1-part-block-0-r-00001-snappy.parquet	parquet	April 26, 2026, 20:47:52 (UTC-04:00)	10.7 MB	Standard

S3 Processed Data – perm_clean/ Folder Contents

The perm_clean/ folder contains 11 Snappy-compressed Parquet files ranging from 1.9 MB to 10.5 MB, reflecting that the larger PERM dataset is split into more partitions by the Glue ETL job than the H-1B output.

The screenshot shows the Amazon S3 console interface for the bucket `visa-processed-data` under the prefix `perm_clean/`. The left sidebar shows the navigation menu with options like Buckets, Files, Access management and security, and Storage management and insights. The main content area displays a table of 11 objects, all of which are Parquet files. The table includes columns for Name, Type, Last modified, Size, and Storage class. The objects are listed in descending order of size.

Name	Type	Last modified	Size	Storage class
<code>run-AmazonS3_node1776283756959-1-part-block-Q+-00000-snappy.parquet</code>	parquet	April 26, 2026, 20:48:01 (UTC-04:00)	10.5 MB	Standard
<code>run-AmazonS3_node1776283756959-1-part-block-Q+-00001-snappy.parquet</code>	parquet	April 26, 2026, 20:48:01 (UTC-04:00)	8.8 MB	Standard
<code>run-AmazonS3_node1776283756959-1-part-block-Q+-00002-snappy.parquet</code>	parquet	April 26, 2026, 20:47:58 (UTC-04:00)	5.9 MB	Standard
<code>run-AmazonS3_node1776283756959-1-part-block-Q+-00003-snappy.parquet</code>	parquet	April 26, 2026, 20:47:58 (UTC-04:00)	5.6 MB	Standard
<code>run-AmazonS3_node1776283756959-1-part-block-Q+-00004-snappy.parquet</code>	parquet	April 26, 2026, 20:48:00 (UTC-04:00)	5.2 MB	Standard
<code>run-AmazonS3_node1776283756959-1-part-block-Q+-00005-snappy.parquet</code>	parquet	April 26, 2026, 20:48:00 (UTC-04:00)	5.6 MB	Standard
<code>run-AmazonS3_node1776283756959-1-part-block-Q+-00006-snappy.parquet</code>	parquet	April 26, 2026, 20:48:00 (UTC-04:00)	6.6 MB	Standard
<code>run-AmazonS3_node1776283756959-1-part-block-Q+-00007-snappy.parquet</code>	parquet	April 26, 2026, 20:48:00 (UTC-04:00)	5.7 MB	Standard
<code>run-AmazonS3_node1776283756959-1-part-block-Q+-00008-snappy.parquet</code>	parquet	April 26, 2026, 20:48:06 (UTC-04:00)	5.6 MB	Standard
<code>run-AmazonS3_node1776283756959-1-part-block-Q+-00009-snappy.parquet</code>	parquet	April 26, 2026, 20:48:07 (UTC-04:00)	6.7 MB	Standard
<code>run-AmazonS3_node1776283756959-1-part-block-Q+-00010-snappy.parquet</code>	parquet	April 26, 2026, 20:47:54 (UTC-04:00)	1.9 MB	Standard

EC2 Configuration & Deployment

EC2 instance was launched with the following configuration:

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

e.g. My Web Server

Add additional tags

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

RedHat

SUSE Linux

SUSE

Debian

debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-098e39bafa7e7303d (64-bit (x86), uefi-preferred) / ami-03ed25db53d8de46c (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.11.20260413.0 x86_64 HVM kernel-6.1

Architecture

64-bit (x86)

Boot mode

uefi-preferred

AMI ID

ami-098e39bafa7e7303d

Publish Date

2026-04-10

Username

ec2-user

Verified provider

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.small

Family: t3 2 vCPU 2 GiB Memory Current generation: true On-Demand RHEL base pricing: 0.0496 USD per Hour
On-Demand Windows base pricing: 0.0392 USD per Hour On-Demand SUSE base pricing: 0.0518 USD per Hour
On-Demand Linux base pricing: 0.0208 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0245 USD per Hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

visa-dashboard-key

Create new key pair

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-03b604fb44ce00df9 (project-vpc)

10.0.0.0/16

Create new VPC

Subnet [Info](#)

subnet-03f8472e6fde5015c

project-subnet-public1-us-east-1a

VPC: vpc-03b604fb44ce00df9 Owner: 718104182325 Availability Zone: us-east-1a (us-east-1a) (us-east-1a) (us-east-1a)
Zone type: Availability Zone IP addresses available: 4089 CIDR: 10.0.0.0/20

Create new subnet

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Common security groups [Info](#)

Select security groups

ec2-sg sg-06b4a5c4fde394add X
VPC: vpc-03b604fb44ce00df9

Compare security group rules

Security groups that you add or remove here will be added to or removed from all your network interfaces.

▼ Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.11....[read more](#)
ami-098e39bafa7e7303d

Virtual server type (instance type)

t3.small

Firewall (security group)

ec2-sg

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

The EC2 security group was configured with the following inbound rules to allow SSH access for deployment and browser access to the Streamlit dashboard:

Security groups						
sg-06b4a5c4fdc394add (ec2-sg)						
Inbound rules						
Filter rules						
Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
-	sgr-0b6c60e046a629942	8501	TCP	0.0.0.0/0	ec2-sg	-
-	sgr-0379224bf280add4c	22	TCP	0.0.0.0/0	ec2-sg	-
-	sgr-06fd8e85275bc6e43	80	TCP	0.0.0.0/0	ec2-sg	-

The EC2 was successfully deployed as per below:

EC2

Dashboard

AWS Global View

Events

Instances

- Instances
- Instance Types
- Launch Templates
- Spot Requests
- Savings Plans
- Reserved Instances
- Dedicated Hosts
- Capacity Reservations
- Capacity Manager

Images

- AMIs
- AMI Catalog

Elastic Block Store

- Volumes
- Snapshots
- Lifecycle Manager

Network & Security

- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs
- Network Interfaces

Instances (1/1)

Last updated 1 minute ago

[Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets
Choose filter set

Find Instance by attribute or tag (case-sensitive)

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPV
☒	visa-dashboar...	i-Oefff728a8426d7aa7	Running	t3.small	3/3 checks passed	View alarms +	us-east-1a	ec2-52-204-37-56.com...	52.204.37.56

i-Oefff728a8426d7aa7 (visa-dashboard-ec2)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

▼ Instance summary

Instance ID i-Oefff728a8426d7aa7	Public IPv4 address 52.204.37.56 open address	Private IPv4 addresses 10.0.7.119
IPv6 address -	Instance state ● Running	Public DNS ec2-52-204-37-56.compute-1.amazonaws.com open address
Hostname type IP name: ip-10-0-7-119.ec2.internal	Private IP DNS name (IPv4 only) ip-10-0-7-119.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.small	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 52.204.37.56 [Public IP]	VPC ID vpc-03b604fb44ce00df9 (project-vpc)	Auto Scaling Group name -
IAM role -	Subnet ID subnet-03f8472e6fde5015c (project-subnet-public1-us-east-1a)	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-1:718104182325:instance/i-Oefff728a8426d7aa7	
Operator -		

Streamlit dashboard

The Streamlit dashboard was deployed on an EC2 instance that has been created.

```
bonirosen — ec2-user@ip-10-0-7-119:~ — ssh -i ~/.ssh/visa-dashboard-key.pem ec2-user@52.204.37...
Last login: Mon Apr 27 13:21:44 on ttys000
(base) bonirosen@Heidis-Laptop ~ % ssh -i ~/.ssh/visa-dashboard-key.pem ec2-user@52.204.37.56
The authenticity of host '52.204.37.56 (52.204.37.56)' can't be established.
ED25519 key fingerprint is SHA256:EqA2wuriRcgZPS1NrH2L1CPH3N9F01z0LtPbCyDct9Q.
This host key is known by the following other names/addresses:
~/.ssh/known_hosts:3: 34.229.106.19
~/.ssh/known_hosts:5: 98.87.5.0
~/.ssh/known_hosts:6: 3.82.56.2
~/.ssh/known_hosts:7: 54.167.208.5
~/.ssh/known_hosts:8: 54.80.134.151
~/.ssh/known_hosts:9: 34.239.138.159
~/.ssh/known_hosts:10: 34.239.49.58
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '52.204.37.56' (ED25519) to the list of known hosts.

      _#_
     _###_   Amazon Linux 2023
    _###_
   _###_
  _###_
 _###_
/###_
V~'  ---> https://aws.amazon.com/linux/amazon-linux-2023
~m/

Last login: Mon Apr 27 04:07:19 2026 from 18.206.107.29
[ec2-user@ip-10-0-7-119 ~]$ cd ~/visa-dashboard
```

After the directory has been created inside EC2, we use nano text editor to create an app.py file to develop the dashboard python code.

```
bonirosen — nano ~/visa-dashboard/app.py — 135x49
..9:~ — ssh -i ~/.ssh/visa-dashboard-key.pem ec2-user@52.204.37.56 ... nano ~/visa-dashboard/app.py
UW PICO 5.09 File: /Users/bonirosen/visa-dashboard/app.py

import streamlit as st
import pandas as pd
import plotly.express as px
import psycopg2

DB_HOST = "visa-group1-postgres.clyuiasiev48.us-east-1.rds.amazonaws.com"
DB_PORT = 5432
DB_NAME = "visa_analytics"
DB_USER = "postgres"
DB_PASS = "postgres"

@st.cache_resource
def get_connection():
    return psycopg2.connect(host=DB_HOST, port=DB_PORT, dbname=DB_NAME, user=DB_USER, password=DB_PASS)

@st.cache_data
def run_query(query, params=None):
    conn = get_connection()
    return pd.read_sql(query, conn, params=params)

st.set_page_config(page_title="U.S. Visa Sponsorship Analytics", layout="wide", initial_sidebar_state="expanded")

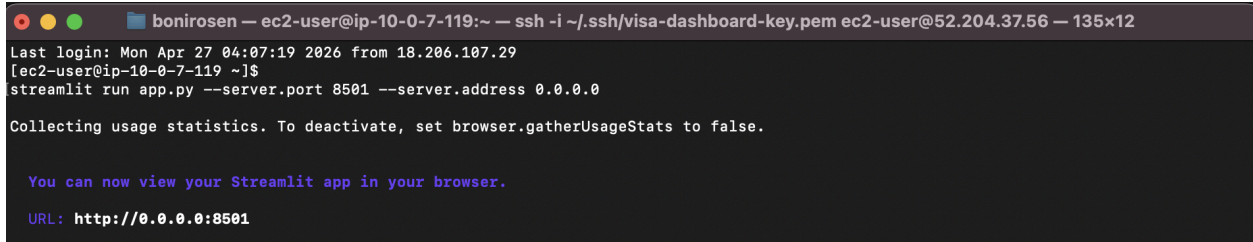
st.markdown("""
<style>
[data-testid="stAppViewContainer"] { background-color: #0f1117; }
[data-testid="stSidebar"] { background-color: #1a1a2e; }
.kpi-card { background: #16213e; border: 1px solid #0f3460; border-radius: 12px; padding: 1.2rem; text-align: center; margin-bottom: 1$
.kpi-value { color: #e94560; font-size: 1.6rem; font-weight: 700; }
.kpi-label { color: #a8b2d8; font-size: 0.8rem; margin-top: 0.3rem; }
.kpi-sublabel { color: #2ecc71; font-size: 0.75rem; margin-top: 0.2rem; }
.section-title { color: #e94560; font-size: 1.1rem; font-weight: 600; border-left: 3px solid #e94560; padding-left: 0.7rem; margin: 1r$
</style>
""", unsafe_allow_html=True)

st.markdown("<h1 style='color:#e94560;text-align:center;padding-top:1rem;'>U.S. Visa Sponsorship Analytics</h1>", unsafe_allow_html=True)
st.markdown("<p style='color:#a8b2d8;text-align:center;'>Explore H-1B and PERM sponsorship trends across employers, states, and years $
st.markdown("<hr>")

st.sidebar.markdown("<h2 style='color:#e94560;'>Filters</h2>", unsafe_allow_html=True)
st.sidebar.markdown("<hr>")
year_range = st.sidebar.slider("Year Range", min_value=2008, max_value=2026, value=(2015, 2026))
st.sidebar.markdown("<hr>")
st.sidebar.markdown("<div style='background:#16213e;border-radius:8px;padding:0.8rem;'><p style='color:#e94560;font-weight:600;margin:$
st.sidebar.markdown("<br><small style='color:#a8b2d8;'>APAN 5450 | Group 1<br>Columbia University SPS</small>", unsafe_allow_html=True)

Get Help      WriteOut      Read File      Prev Pg      Cut Text      Cur Pos
Exit          Justify        Where is       Next Pg      UnCut Text    To Spell
```

We launched the Streamlit dashboard once app.py has been successfully created and copied to EC2 (visa-dashboard).

A terminal window with a dark background and light text. The title bar shows a macOS-style window with red, yellow, and green buttons, and the text 'bonirosen — ec2-user@ip-10-0-7-119:~ — ssh -i ~/.ssh/visa-dashboard-key.pem ec2-user@52.204.37.56 — 135x12'. The terminal content shows a successful login, a prompt, the execution of 'streamlit run app.py --server.port 8501 --server.address 0.0.0.0', a message about usage statistics, a confirmation message in purple, and the URL 'http://0.0.0.0:8501' in blue.

```
bonirosen — ec2-user@ip-10-0-7-119:~ — ssh -i ~/.ssh/visa-dashboard-key.pem ec2-user@52.204.37.56 — 135x12
Last login: Mon Apr 27 04:07:19 2026 from 18.206.107.29
[ec2-user@ip-10-0-7-119 ~]$
streamlit run app.py --server.port 8501 --server.address 0.0.0.0

Collecting usage statistics. To deactivate, set browser.gatherUsageStats to false.

You can now view your Streamlit app in your browser.

URL: http://0.0.0.0:8501
```

Glue (ETL)

h1b-clean-job

We configured the h1b-clean-job using the visual editor to transform raw H-1B data. Key attributes include the source node (S3 raw dataset), transformation steps (data cleaning/mapping), and the target node, which writes to S3 in Parquet format, enabling efficient downstream querying.

The screenshot displays the AWS Glue console interface for the 'h1b-clean-job'. The top navigation bar includes tabs for 'Visual', 'Script', 'Job details', 'Runs', 'Data quality', 'Schedules', and 'Version Control'. The 'Visual' tab is active, showing a workflow diagram with a single node labeled 'Data source - Data Catalog' and 'AWS Glue Data Catalog'. The right-hand panel shows the 'Data source properties - Data Catalog' configuration, including fields for 'Name' (AWS Glue Data Catalog), 'Database' (visapath_glue_db), 'Table' (h1b_h1b), and a 'Partition predicate' section. The 'Data preview' section at the bottom left indicates 'Loading data previews. (0% complete)'. The 'Output schema' section at the bottom right shows 'Previewing 0 of 0 fields'.

Data flow within the Glue job showing extraction from S3, transformation logic, and schema mapping. Important attributes include column transformations and data preview, which validate that fields are correctly cleaned and standardized before output.

The screenshot displays the AWS Glue console interface for the 'h1b-clean-job'. The top navigation bar includes tabs for 'Visual', 'Script', 'Job details', 'Runs', 'Data quality', 'Schedules', and 'Version Control'. The 'Visual' tab is active, showing a workflow diagram with two nodes: 'Transform - Change Schema' and 'Transform - SQL Query'. The right-hand panel shows the 'Transform' configuration for the 'Change Schema' node, including a 'Change Schema (Apply mapping)' table. The 'Data preview' section at the bottom left indicates 'Loading data previews. (0% complete)'. The 'Output schema' section at the bottom right shows 'Previewing 0 of 0 fields'.

Source key	Target key	Data type	Drop
line by line			☒
fiscal year	fiscal_year	long	☐
employer (petition	employer_	string	☐
tax id	tax_id	string	☐
industry (naics) co	industry_n	string	☐
petitioner city	employer_	string	☐
petitioner state	employer_	string	☐
petitioner zip code	employer_	string	☐
new employment :	new_empl	long	☐
new employment +	new_empl	long	☐
continuation appri	continuat	long	☐
continuation denis	continuat	long	☐
change with same	change_sa	long	☐
change with same	change_sa	long	☐

h1b-clean-job (Last modified on 4/26/2026, 8:15:48 PM)

Visual | Script | Job details | Runs | Data quality | Schedules | Version Control

Transform

Name: SQL Query

Node parents: Choose one or more parent node

Change Schema (X) ApplyMapping - Transform

Associate an alias with each input source: Info

Input sources: Change Schema SQL aliases: input

SQL query: Enter a SQL statement to add to your job.

```

1 SELECT
2   fiscal_year,
3
4   -- Employee name: trim + uppercase + remove punctuation
5   REGEXP_REPLACE(
6     UPPER(TRIM(employer_name)),
7     '["\s-@&9 ]',
8     ''
9   ) AS employer_name,
10
11   tax_id,
12   industry_naics_code

```

Data preview (200) (Info) (READY) (End session) (Previewing 20 of 20 fields)

fiscal_year	employer_name	tax_id	industry_naics_code	employer_city
2009	ZYLOG SYSTEMS LTD DBA ZSL INC	6210.0	54 - Professional, Scientific, and Technical Services	EDISON
2009	ZYLOG SYSTEMS LTD	6210.0	54 - Professional, Scientific, and Technical Services	EDISON

h1b-clean-job (Last modified on 4/26/2026, 8:42:14 PM)

Visual | Script | Job details | Runs | Data quality | Schedules | Version Control

Change Schema

Transform - SQL Query SQL Query

Data target properties - S3

Name: Amazon S3

Node parents: Choose one or more parent node

SQL Query (X) SqlCode - Transform

Format: Parquet

Compression Type: Snappy

S3 Target Location

Choose an S3 location in the format s3://bucket/prefix/object/ with a trailing slash (/).

s3://visa-processed-data/h1b_clean/ (View) (Browse S3)

Data Catalog update options (Info)

Choose how you want to update the Data Catalog table's schema and partitions. These options will only apply if the Data Catalog table is an S3 backed source.

☒ Do not update the Data Catalog

☐ Create a table in the Data Catalog and on subsequent runs, update the schema and add new partitions

☐ Create a table in the Data Catalog and on subsequent runs, keep existing schema and add new partitions

Perm-clean-job

We configured perm-clean-job to process PERM labor certification data. Key attributes include the S3 source dataset, transformation steps, and S3 target output, demonstrating parallel processing of a second data pipeline.

perm-clean-job

Last modified on 4/26/2026, 8:45:47 PM Actions Save Run

Visual Script Job details Runs Data quality Schedules Version Control

Visual

Data source - Data Catalog
AWS Glue Data Catalog

Data preview Info STARTING (2%) ⓘ

Filter sample dataset

End session Previewing 0 of 0 fields

Starting up data preview. This may take a few seconds.

Data source properties - Data Catalog

Name
AWS Glue Data Catalog

Database
Choose a database.
visapath_glue_db

Use runtime parameters

Table
perm_perm

Use runtime parameters

Partition predicate - optional
Enter a boolean expression supported by Spark SQL, using only partition columns.

Partition predicate syntax for Spark SQL is year == year(date_sub(current_date, 7)) AND month == month(date_sub(current_date, 7)) AND day == day(date_sub(current_date, 7)).

Transformation pipeline for PERM data, showing source → transform → target flow. Important attributes include schema changes and partitioned output configuration, which explain why multiple Parquet files are generated for larger datasets.

perm-clean-job

Last modified on 4/26/2026, 8:45:47 PM Actions Save Run

Visual Script Job details Runs Data quality Schedules Version Control

Visual

Transform - Change Schema
Change Schema

Transform - SQL Query
SQL Query

Data preview Info STARTING (0%) ⓘ

Filter sample dataset

End session Previewing 0 of 0 fields

Starting up data preview. This may take a few seconds.

Transform

Name
Change Schema

Node parents
Choose which nodes will provide inputs for this one.
Choose one or more parent node
AWS Glue Data Catalog X
Catalog - DataSource

Change Schema (Apply mapping)

Source key	Target key	Data type	Drop
case_number	case_num	string	☐
case_status	case_status	string	☐
decision_date	decision_d	string	☐
received_date	received_d	string	☐
application_type	application	string	☐
refile			☒
orig_file_date			☒
employer_name	employer_	string	☐
employer_address	employer_	string	☐
employer_city	employer_	string	☐
employer_state	employer_	string	☐
employer_postal_s	employer_	string	☐
employer_country	employer_	string	☐
employer_phone			☒
employer_num en			☒

perm-clean-job Last modified on 4/26/2026, 8:45:47 PM

Visual | Script | Job details | Runs | Data quality | Schedules | Version Control

Data target properties - S3

Name: Amazon S3

Node parents: Choose one or more parent node

Format: Parquet

Compression Type: Snappy

S3 Target Location: Choose an S3 location in the format s3://bucket/prefix/object/ with a trailing slash (/).
s3://visa-processed-data/perm_c

Data Catalog update options: Do not update the Data Catalog

Data preview | Output schema

Target node not supported
You have selected a data target node which is not supported for data preview. Please select another type of node instead.

perm-clean-job Last modified on 4/26/2026, 8:45:47 PM

Visual | Script | Job details | Runs | Data quality | Schedules | Version Control

Data preview | Output schema

Data preview (200) **READY** **End session** **Previewing 27 of 27 fields**

case_number	application_type	decision_date	filing_year	case_status
A-07204-59294	PERM	2007-11-15	2007	CERTIFIED-EXPIRED
C-07276-81772	PERM	2007-10-12	2007	DENIED

Transform

Name: SQL Query

Node parents: Choose one or more parent node

Change Schema

Associate an alias with each input source

Input sources: Change Schema

SQL aliases: Input

SQL query

```
1 SELECT
2   case_number,
3   application_type,
4   decision_date,
5   YEAR(CAST(decision_date AS DATE)) AS filing_year,
6   case_status,
7   REGEXP_REPLACE(
8     UPPER(TRIM(employer_name)),
9     '[^A-Z0-9 ]',
10    ''
11  )
12  FROM ...
```

test-s3-glue-job

Script-based Glue job (test-s3-glue-job) showing the underlying PySpark code used for ETL. Key attributes include the script logic (read from S3, transform data, write output) and job execution environment, highlighting custom processing beyond the visual editor.

test-s3-glue-job

Last modified on 4/25/2026, 10:50:53 AM

Actions

Save

Run

Script

Job details

Runs

Data quality

Schedules

Version Control

Script

```
1 import sys
2 from aws glue.transforms import *
3 from aws glue.utils import getResolvedOptions
4 from pyspark.context import SparkContext
5 from aws glue.context import GlueContext
6 from aws glue.job import Job
7
8 args = getResolvedOptions(sys.argv, ['JOB_NAME'])
9 sc = SparkContext()
10 glueContext = GlueContext(sc)
11 spark = glueContext.spark_session
12 job = Job(glueContext)
13 job.init(args['JOB_NAME'], args)
14
15 # RDS connection URL
16 JDBC_URL = "jdbc:postgresql://visa-group1-postgres.clyluasiev48.us-east-1.rds.amazonaws.com:5432/visa_analytics"
17 JDBC_PROPS = {
18     "user": "postgres",
19     "password": "postgres",
20     "driver": "org.postgresql.Driver"
21 }
22
23 # -- HIB: S3 processed -> RDS hib_sponsors --
24 hib_df = spark.read.parquet("s3://visa-processed-data/hib_clean/")
25
26 hib_df.write \
27     .format("jdbc") \
```

Python Ln 1, Col 1 Errors: 0 Warnings: 0

DB Subnet

aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 7181-0418-2325
voclabs/user4935500=Group_1

Aurora and RDS

Subnet groups

Create DB subnet group

Create DB subnet group

To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.

Subnet group details

Name
You won't be able to modify the name after your subnet group has been created.

Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

Description

VPC
Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.

project-vpc (vpc-03b604fb44ce00df9)
4 Subnets, 2 Availability Zones

Add subnets

Availability Zones
Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone

us-east-1a X us-east-1b X

Subnets
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets

project-subnet-private1-us-east-1a X
Subnet ID: subnet-082eb046c0085cff4 CIDR: 10.0.128.0/20

project-subnet-private2-us-east-1b X
Subnet ID: subnet-0fe5e7a7cd4797fa9 CIDR: 10.0.144.0/20

For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.

Availability zone	Subnet name	Subnet ID	CIDR block
us-east-1a	project-subnet-private1-us-east-1a	subnet-082eb046c0085cff4	10.0.128.0/20
us-east-1b	project-subnet-private2-us-east-1b	subnet-0fe5e7a7cd4797fa9	10.0.144.0/20

Cancel

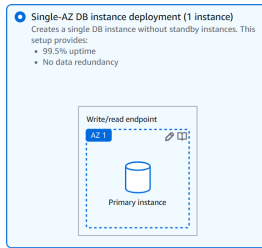
Create

DB Instance > Create database > Full configuration (RDS)

Engine Options: PostgreSQL

Database Creation Method: Full Configuration

Templates: Dev/Test



DB instance identifier: visa-group1-postgres

Master username: postgres

Credentials Management: Self-managed

Master password: postgres

DB instance class: Burstable classes (includes t classes)

Instance type: db.t3.micro

Storage type: General Purpose SSD (gp3)

Allocated storage: 20

Additional storage configuration > storage autoscaling > disabled storage autoscaling

Connectivity [Info](#)

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ **Don't connect to an EC2 compute resource**
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ **Connect to an EC2 compute resource**
Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) [Info](#)
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

project-vpc (vpc-03b604fb44ce00df9)
4 Subnets, 2 Availability Zones

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

DB subnet group [Info](#)
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

group1-rds-subnet-group
2 Subnets, 2 Availability Zones

Public access [Info](#)

☐ **Yes**
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☒ **No**
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ **Choose existing**
Choose existing VPC security groups

☐ **Create new**
Create new VPC security group

Existing VPC security groups

Choose one or more options

rds-sg X

Availability Zone [Info](#)

No preference

RDS Proxy
RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

☐ **Create an RDS Proxy** [Info](#)
RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

▼ Additional configuration

Database options, encryption turned on, backup turned off, backtrack turned off, maintenance, CloudWatch Logs, delete protection turned off.

Database options

Initial database name [Info](#)

If you do not specify a database name, Amazon RDS does not create a database.

DB parameter group [Info](#)

Option group [Info](#)

Backup

☐ Enable automated backup

Creates a point-in-time snapshot of your database

Backup tags [Info](#)

Copies tags from the source database to the automated backup and snapshots respectively.

☐ Copy tags to automated backup

This is a one-time setting. Future tag modifications need manual updates.

☒ Enable encryption

Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console. [Info](#)

AWS KMS key [Info](#)

Account

718104182325

KMS key ID

alias/aws/rds

Maintenance

Auto minor version upgrade [Info](#)

☒ Enable auto minor version upgrade

Enabling auto minor version upgrade will automatically upgrade your database minor version. For limitations and more details, see [Automatically upgrading the minor engine version documentation](#).

Maintenance window [Info](#)

Select the period you want pending modifications or maintenance applied to the database by Amazon RDS.

☐ Choose a window

☒ No preference

☐ Enable deletion protection

Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.

Glue Trigger

We implemented a scheduled Glue trigger to automate the ETL pipeline, which processes data from S3 and loads it into RDS every month

AWS Glue > Triggers > Add trigger

Glue

ng started

obs

sual ETL

otebooks

ib run monitoring

Catalog tables

connections

.flows (orchestration)

ETL integrations [New](#)

Catalog

Integration and ETL

ETL integrations [New](#)

obs

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ib run monitoring

active Sessions

classification tools

nsitive data detection

scord Matching

iers

.flows (orchestration)

ueprints

urity configurations

manaoement [New](#)

Set trigger properties

Step 2: Choose jobs or crawlers to activate

Step 3: Review and create

Set trigger properties

Trigger details

Name

monthly-glue-trigger

Name can be up to 255 characters long. Some character set including control characters are prohibited.

Description - optional

Enter a description

Descriptions can be up to 2048 characters long.

Trigger type

☐ On demand
Fire the trigger immediately when started.

☒ Schedule
Fire the trigger on a timer.

☐ Job or crawler event
Fire the trigger when job or crawler events match your watched list.

Schedule

You can define a time-based schedule for your crawlers and jobs in AWS Glue. The definition of these schedules uses the Unix-like [cron](#) syntax. [Learn more](#).

Frequency

Custom (cron expression)

Schedule

You can define a time-based schedule for your crawlers and jobs in AWS Glue. The definition of these schedules uses the Unix-like [cron](#) syntax. [Learn more](#).

Frequency

Custom (cron expression)

0

0

1

*

?

*

MinutesHoursDay of monthMonthDay of weekYear

Generated result: cron(0 0 1 * ? *)

At 12:00 AM, on day 1 of the month

Tags - optional

Use tags to organize and identify your resources.

Cancel

Next

Add target



Resource type

Job

The type of resource for this trigger target

Job name

test-s3-glue-job

Job to start when this trigger fires

► Parameters passed down to job "test-s3-glue-job" when started - optional

Cancel

Add

Trigger successfully created

The following trigger was created: "monthly-glue-trigger"

monthly-glue-trigger

Last updated (UTC)
April 20, 2026 at 03:19:02

Edit trigger

Trigger properties

Name monthly-glue-trigger	Description -
Trigger type Scheduled	Status Created
Associated workflow -	Schedule At 12:00 AM, on day 1 of the month

Target resources

Tags

Resources to trigger (1)

List of resources to start once the trigger activates

Type	Name	Parameters
Job	test-s3-glue-job	-

VPC

Certificate authority - *optional* [Info](#)

Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rdc-ca-rsa2048-g1 (default)
Expiry: May 25, 2061

If you don't select a certificate authority, RDS chooses one for you.

▼ Additional configuration

Database port [Info](#)

TCP/IP port that the database will use for application connections.

5432

Tags - *optional*

A tag consists of a case-sensitive key-value pair.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Monitoring [Info](#)

Choose monitoring tools for this database. Database Insights provides a combined view of Performance Insights and Enhanced Monitoring for your fleet of databases. **Database Insights** pricing is separate from RDS monthly estimates. See [Amazon CloudWatch pricing](#).

☐ Database Insights - Advanced

- Retains 15 months of performance history
- Fleet-level monitoring
- Integration with CloudWatch Application Signals

☒ Database Insights - Standard

- Retains 7 days of performance history, with the option to pay for the retention of up to 24 months of performance history

Performance Insights

☐ Enable Performance Insights

With Performance Insights dashboard, you can visualize the database load on your Amazon RDS DB instance load and filter the load by waits, SQL statements, hosts, or users.

▼ Additional monitoring settings

Enhanced Monitoring, CloudWatch Logs and DevOps Guru

Enhanced Monitoring

☐ Enable Enhanced monitoring

Enabling Enhanced Monitoring metrics are useful when you want to see how different processes or threads use the CPU.

Log exports

Select the log types to publish to Amazon CloudWatch Logs

- ☐ iam-db-auth-error log
- ☐ PostgreSQL log
- ☐ Upgrade log

We designed a VPC with separate public and private subnets across multiple availability zones. The public subnet is connected to the Internet Gateway via a route table entry (0.0.0.0/0), enabling external access while maintaining isolation for backend resources.

Details

VPC ID

vpc-03b604fb44ce00df9

DNS resolution

Enabled

Main network ACL

acl-0822454ad3bfb4440

IPv6 CIDR (Network border group)

-

Encryption control ID

-

State

Available

Tenancy

default

Default VPC

No

Network Address Usage metrics

Disabled

Encryption control mode

-

Block Public Access

Off

DHCP option set

dopt-0ed5cc0a497144c9d

IPv4 CIDR

10.0.0.0/16

Route 53 Resolver DNS Firewall rule groups

Failed to load rule groups

DNS hostnames

Enabled

Main route table

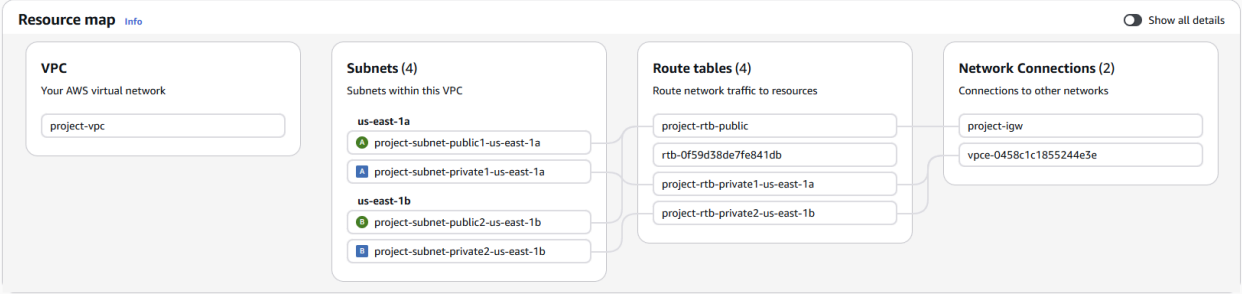
rtb-0f59d38de7fe841db

IPv6 pool

-

Owner ID

718104182325



Subnets

1) Public

Subnets (1/10) Info

Last updated 6 minutes ago

Actions

Find subnets by attribute or tag

	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR
☒	project-subnet-public2-us-east-1b	subnet-0ef38ceb62c4384ea	Available	vpc-03b604fb44ce00df9 proje...	Off	10.0.16.0/20	-
☐	project-subnet-public1-us-east-1a	subnet-03f8472e6fde5015c	Available	vpc-03b604fb44ce00df9 proje...	Off	10.0.0.0/20	-
☐	project-subnet-private2-us-east-1b	subnet-0fe5e7a7cd4797fa9	Available	vpc-03b604fb44ce00df9 proje...	Off	10.0.144.0/20	-
☐	project-subnet-private1-us-east-1a	subnet-082eb046c0085cff4	Available	vpc-03b604fb44ce00df9 proje...	Off	10.0.128.0/20	-
☐	-	subnet-06b1e9887f371bfd7	Available	vpc-0ef75dbfb49a48021	Off	172.31.0.0/20	-
☐	-	subnet-0dc023774e9fa0c36	Available	vpc-0ef75dbfb49a48021	Off	172.31.16.0/20	-
☐	-	subnet-0bb46374a0a726956	Available	vpc-0ef75dbfb49a48021	Off	172.31.48.0/20	-
☐	-	subnet-0ec9b6683293dd8e3	Available	vpc-0ef75dbfb49a48021	Off	172.31.64.0/20	-
☐	-	subnet-09541b464701ecf70	Available	vpc-0ef75dbfb49a48021	Off	172.31.32.0/20	-
☐	-	subnet-0fefe021fd73af2f9	Available	vpc-0ef75dbfb49a48021	Off	172.31.80.0/20	-

subnet-0ef38ceb62c4384ea / project-subnet-public2-us-east-1b

Route table: rtb-00b7c13be2d790a19 / project-rtb-public

Edit route table

Routes (2)

Filter routes

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	igw-087a58cb74a3a6ed9

2) Private

Subnets (1/10) Info

Last updated 6 minutes ago

Actions

Create subnet

Find subnets by attribute or tag

	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR
☐	project-subnet-public2-us-east-1b	subnet-0ef38ceb62c4384ea	Available	vpc-03b604fb44ce00df9 proje...	Off	10.0.16.0/20	-
☐	project-subnet-public1-us-east-1a	subnet-03f8472e6fde5015c	Available	vpc-03b604fb44ce00df9 proje...	Off	10.0.0.0/20	-
☒	project-subnet-private2-us-east-1b	subnet-0fe5e7a7cd4797fa9	Available	vpc-03b604fb44ce00df9 proje...	Off	10.0.144.0/20	-
☐	project-subnet-private1-us-east-1a	subnet-082eb046c0085cff4	Available	vpc-03b604fb44ce00df9 proje...	Off	10.0.128.0/20	-
☐	-	subnet-06b1e9887f371bfd7	Available	vpc-0ef75dbfb49a48021	Off	172.31.0.0/20	-
☐	-	subnet-0dc023774e9fa0c36	Available	vpc-0ef75dbfb49a48021	Off	172.31.16.0/20	-
☐	-	subnet-0bb46374a0a726956	Available	vpc-0ef75dbfb49a48021	Off	172.31.48.0/20	-
☐	-	subnet-0ec9b6683293dd8e3	Available	vpc-0ef75dbfb49a48021	Off	172.31.64.0/20	-
☐	-	subnet-09541b464701ecf70	Available	vpc-0ef75dbfb49a48021	Off	172.31.32.0/20	-
☐	-	subnet-0fefe021fd73af2f9	Available	vpc-0ef75dbfb49a48021	Off	172.31.80.0/20	-

subnet-0fe5e7a7cd4797fa9 / project-subnet-private2-us-east-1b

Route table: rtb-0b01b5896e80c7826 / project-rtb-private2-us-east-1b

Edit route table association

Routes (2)

Filter routes

Destination	Target
10.0.0.0/16	local
pl-63a5400a	vpce-0458c1c1855244e3e

Security groups lists

Security groups were configured to separate access layers.

EC2 allows public access for web traffic, while RDS restricts access to only EC2 and Glue, ensuring secure communication between application and data layers.

Security Groups (6) Info

Find security groups by attribute or tag

☐	Name	Security group ID	Security group name	VPC ID	Description	Owner
☐	-	sg-02d7880bdf378ee46	alb-sg	vpc-03b604fb44ce00df9	Allow alb	718104182325
☐	-	sg-035057cea531dc54a	rds-sg	vpc-03b604fb44ce00df9	RDS access	718104182325
☐	-	sg-039c9cd41f8daa412	default	vpc-0ef75d1bf49a48021	default VPC security group	718104182325
☐	-	sg-06b4a5c4fdc394add	ec2-sg	vpc-03b604fb44ce00df9	EC2 access	718104182325
☐	-	sg-08040eb18776313ac	glue-sg	vpc-03b604fb44ce00df9	Glue access to S3 and RDS	718104182325
☐	-	sg-0955b7fc99fe74763	default	vpc-03b604fb44ce00df9	default VPC security group	718104182325

EC2 security group

sg-06b4a5c4fdc394add - ec2-sg

Actions

Details

Security group name

ec2-sg

Owner

718104182325

Security group ID

sg-06b4a5c4fdc394add

Inbound rules count

3 Permission entries

Description

EC2 access

Outbound rules count

1 Permission entry

VPC ID

vpc-03b604fb44ce00df9

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (3)

Search

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-0b6c60e046a629942	IPv4	Custom TCP	TCP	8501	0.0.0.0/0	-
☐	-	sgr-0379224bf280add4c	IPv4	SSH	TCP	22	0.0.0.0/0	-
☐	-	sgr-06fd8e85275bc6e43	IPv4	HTTP	TCP	80	0.0.0.0/0	-

RDS SG

sg-035057cea531dc54a - rds-sg

Actions

Details

Security group name

rds-sg

Owner

718104182325

Security group ID

sg-035057cea531dc54a

Inbound rules count

3 Permission entries

Description

RDS access

Outbound rules count

1 Permission entry

VPC ID

vpc-03b604fb44ce00df9

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (3)

Search

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-06198fc7a0151d724	-	PostgreSQL	TCP	5432	sg-0c1bf8e206ad74116...	-
☐	-	sgr-06ba5c2b9dc265908	-	PostgreSQL	TCP	5432	sg-08040eb18776313a...	-
☐	-	sgr-03f62c1fb9f910411	-	PostgreSQL	TCP	5432	sg-06b4a5c4fdc394add...	-

Glue SG

sg-08040eb18776313ac - glue-sg

Actions

Details

Security group name

glue-sg

Security group ID

sg-08040eb18776313ac

Description

Glue access to S3 and RDS

VPC ID

vpc-03b604fb44ce00df9

Owner

718104182325

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Search

Manage tags

Edit inbound rules

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-0b9dcaa482556357	-	All TCP	TCP	0 - 65535	sg-08040eb18776313a...	-

ALB SG

sg-02d7880b4f378ee46 - alb-sg

Actions

Details

Security group name

alb-sg

Security group ID

sg-02d7880b4f378ee46

Description

Allow alb

VPC ID

vpc-03b604fb44ce00df9

Owner

718104182325

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Search

Manage tags

Edit inbound rules

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-036f67fd310ebe84f	IPv4	HTTP	TCP	80	0.0.0.0/0	-

ALB Target Group

EC2 > Target groups > Create target group

Create target group

Step 1

Create target group

Step 2 - recommended

Register targets

Step 3

Review and create

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation."

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

Instances

Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.

Suitable for: ALB NLB GWLB

Lambda function

Supports load balancing to a single Lambda function. ALB required as traffic source.

Suitable for: ALB

IP addresses

Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.

Suitable for: ALB NLB GWLB

Application Load Balancer

Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.

Suitable for: NLB

Target group name

Name must be unique per Region per AWS account.

visapath-streamlit-tg

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 21/32

Protocol

Protocol for communication between the load balancer and targets.

HTTP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

8501

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

IPv4

EC2 > Target groups > Create target group

IP address type

Only targets with the indicated IP address type can be registered to this target group.

IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-03b604fb44ce00df9 (project-vpc)

10.0.0.0/16

Create VPC

Protocol version

HTTP1

Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

HTTP2

Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

gRPC

Send requests to targets using gRPC. Supported when the request protocol is gRPC.

Health checks

The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol

HTTP

Health check path

Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.

/

Up to 1024 characters allowed.

EC2 > Target groups > Create target group

Step 1

Create target group

Step 2 - recommended

Register targets

Step 3

Review and create

Register targets - recommended

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (1)

Filter instances

Instance ID	Name	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
i-0df728b8426d7aa7	visa-dashboard-ec2	Running	ec2-sg	us-east-1a	10.0.7.119	subnet-03f8472ed6dc5015c	April 27, 2026, 02:37 (UTC-04:00)

0 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

8501

1-65535 (separate multiple ports with comma)

Include as pending below

1 selection is now pending below. Include more or register targets when ready.

Review targets

Targets (1)

Filter targets

Show only pending

Remove all pending

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
i-0df728b8426d7aa7	visa-dashboard-ec2	8501	Running	ec2-sg	us-east-1a	10.0.7.119	subnet-03f8472ed6dc5015c	April 27, 2026, 02:37 (UTC-04:00)

1 pending

Cancel Previous Next

EC2 > Target groups

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Load Balancers

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Trust Stores

Auto Scaling

Auto Scaling Groups

Settings

Target groups (1/1)

Filter target groups

Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
visapath-streamlit-tg	arn:aws:elasticloadbalancing:us-east-1:718104182325:targetgroup/visapath-streamlit-tg/6d0bd35896b1ed85	8501	HTTP	Instance	visapath-alb	vpc-03b604fb44ce00df9

Target group: visapath-streamlit-tg

Details

Targets

Monitoring

Health checks

Attributes

Tags

Details

arn:aws:elasticloadbalancing:us-east-1:718104182325:targetgroup/visapath-streamlit-tg/6d0bd35896b1ed85

Target type

Instance

Protocol : Port

HTTP: 8501

Protocol version

HTTP1

VPC

vpc-03b604fb44ce00df9

IP address type

IPv4

Load balancer

visapath-alb

1

0

1

0

0

0

Total targets

Healthy

Unhealthy

Unused

Initial

Draining

0 Anomalous

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

ALB

EC2 > Load balancers > Create Application Load Balancer

Create Application Load Balancer

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.

How Application Load Balancers work

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

visapath-alb

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme

Internet-facing

Serves internet-facing traffic.
Has public IP addresses.
DNS name resolves to public IPs.
Requires a public subnet.

Internal

Serves internal traffic.
Has private IP addresses.
DNS name resolves to private IPs.
Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

IPv4

Includes only IPv4 addresses.

Dualstack

Includes IPv4 and IPv6 addresses.

Dualstack without public IPv4

Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with Internet-facing load balancers only.

Network mapping

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view target groups.

vpc-03b604fb44ce00df9 (project-vpc)

Create VPC

IP pools

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the Amazon VPC IP Address Manager console.

Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

us-east-1a (use1-az6)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-03f8472e6fde5015c (project-subnet-public1-us-east-1a)

us-east-1b (use1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-0ef38ce62c4384ea (project-subnet-public2-us-east-1b)

Security groups

A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.

Security groups

Select up to 5 security groups

alb-sg (sg-02d7880b4f378ee46)

Create security group

Listeners and routing

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener HTTP:80

Remove

Protocol HTTP

Port 80

Default action

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

visapath-streamlit-tg

Weight 1

Percent 100%

Add target group

**Note: the target group selected was visapath-streamlit-tg; this is not reflected here, as this is a recreation of the steps to create the ALB.

EC2 > Load balancers

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Target Groups
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▼ **Auto Scaling**
Auto Scaling Groups

Settings

Load balancers (1/1) [What's new?](#)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Q Filter load balancers

☒	Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones	Security groups	DNS name
☒	visapath-alb	Active	application	Internet-facing	IPv4	vpc-03b604fb44ce00df9	2 Availability Zones	sg-02d7880b4f378ee4...	visapath-alb-17

Load balancer: visapath-alb

[Details](#) | [Listeners and rules](#) | [Network mapping](#) | [Resource map](#) | [Security](#) | [Monitoring](#) | [Integrations](#) | [Attributes](#) | [Capacity](#) | [Tags](#)

Details
Load balancer type
Application
Scheme
Internet-facing
Load balancer ARN
arn:aws:elasticloadbalancing:us-east-1:718104182325:loadbalancer/app/visapath-alb/f33c0c387d613f14

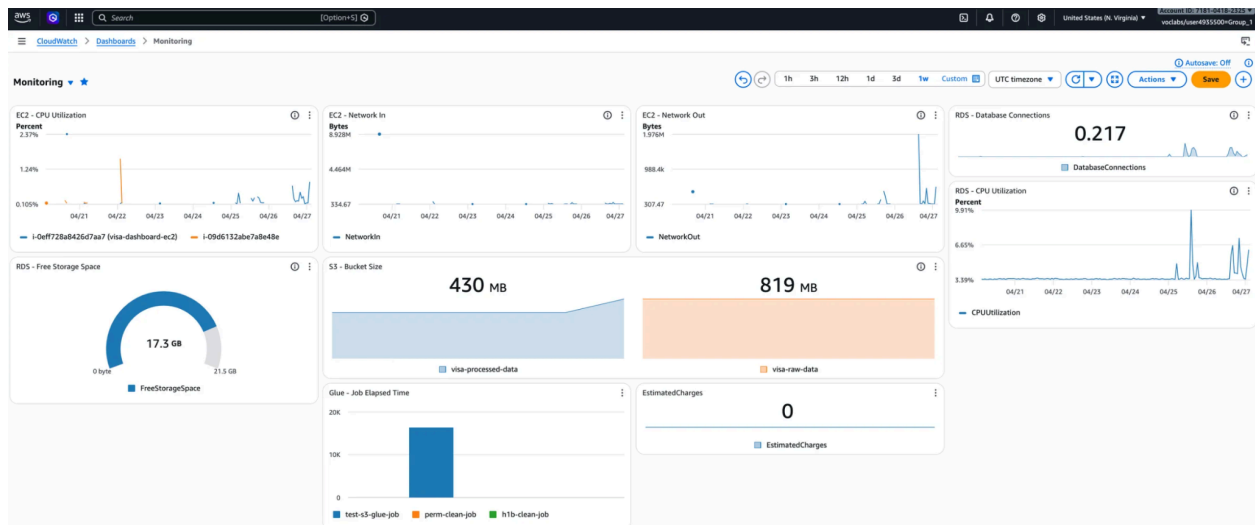
Status
Active
Hosted zone
Z35SXDOTRQ7X7K
DNS name info
visapath-alb-174640799.us-east-1.elb.amazonaws.com (A Record)

VPC
vpc-03b604fb44ce00df9
Availability Zones
 subnet-03f8472e6fde5015c us-east-1a (use1-az6)
 subnet-0ef58ceb62c4384ea us-east-1b (use1-az1)

Load balancer IP address type
IPv4
Date created
April 26, 2026, 21:34 (UTC-04:00)

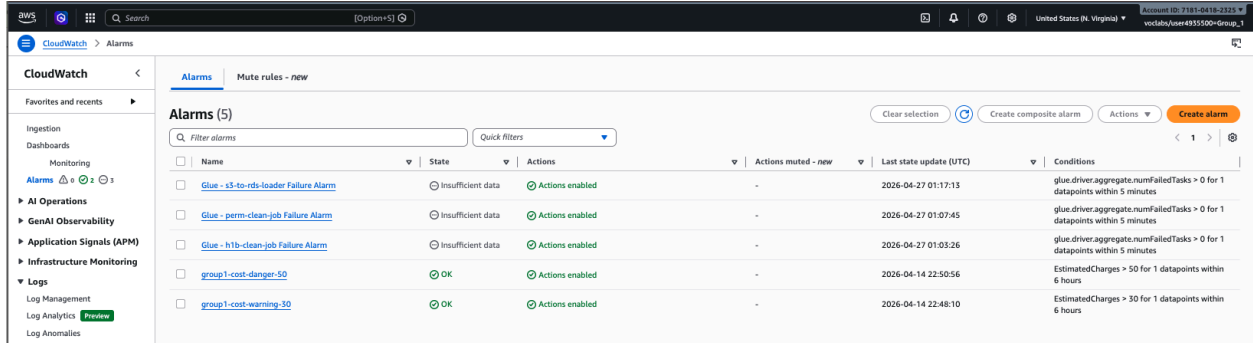
CloudWatch Custom Monitoring Dashboard

A custom CloudWatch dashboard consolidating key metrics across all project services, EC2 CPU/network usage, RDS connections and CPU utilization, RDS free storage, S3 bucket sizes, Glue job elapsed time, and estimated AWS charges, providing a unified view of the entire pipeline's health.



CloudWatch Alarms Overview

Five CloudWatch alarms are configured, three monitoring Glue job failures (s3-to-rds-loader, perm-clean-job, h1b-clean-job) triggered when numFailedTasks > 0 within 5 minutes, and two cost alarms (group1-cost-warning-30 and group1-cost-danger-50) alerting when estimated charges exceed \$30 and \$50, respectively, both currently in OK state.

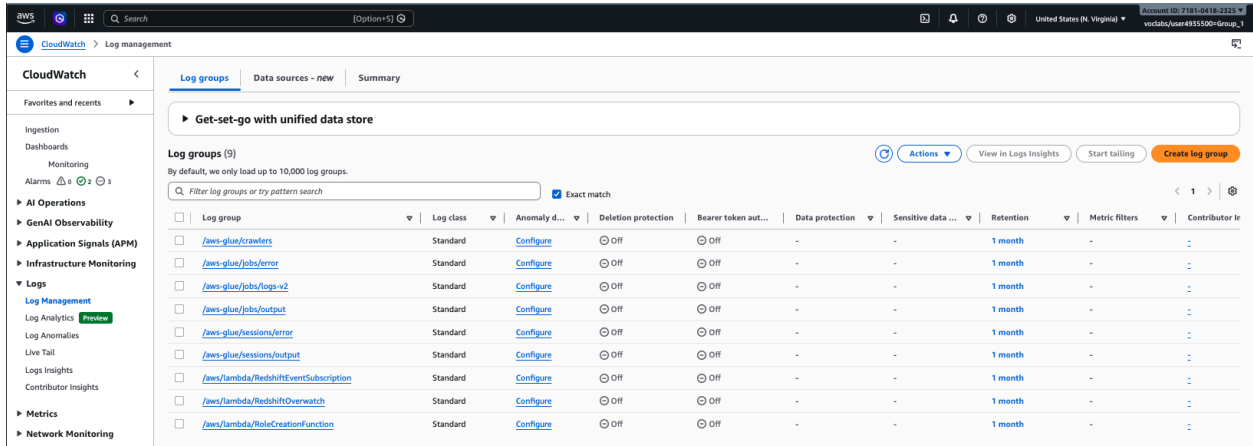


The screenshot shows the AWS CloudWatch Alarms console. The left sidebar contains navigation links for CloudWatch, Favorites and recents, Ingestion, Dashboards, Monitoring, Alarms (5), AI Operations, GenAI Observability, Application Signals (APM), Infrastructure Monitoring, Logs, Log Management, Log Analytics, and Log Anomalies. The main content area is titled 'Alarms (5)' and includes a search bar, 'Quick filters', and buttons for 'Clear selection', 'Create composite alarm', 'Actions', and 'Create alarm'. Below is a table of the five alarms.

☐	Name	State	Actions	Actions muted - new	Last state update (UTC)	Conditions
☐	Glue - s3-to-rds-loader Failure Alarm	Insufficient data	Actions enabled	-	2026-04-27 01:17:13	glue.driver.aggregate.numFailedTasks > 0 for 1 datapoints within 5 minutes
☐	Glue - perm-clean-job Failure Alarm	Insufficient data	Actions enabled	-	2026-04-27 01:07:45	glue.driver.aggregate.numFailedTasks > 0 for 1 datapoints within 5 minutes
☐	Glue - h1b-clean-job Failure Alarm	Insufficient data	Actions enabled	-	2026-04-27 01:03:26	glue.driver.aggregate.numFailedTasks > 0 for 1 datapoints within 5 minutes
☐	group1-cost-danger-50	OK	Actions enabled	-	2026-04-14 22:50:56	EstimatedCharges > 50 for 1 datapoints within 6 hours
☐	group1-cost-warning-30	OK	Actions enabled	-	2026-04-14 22:48:10	EstimatedCharges > 30 for 1 datapoints within 6 hours

CloudWatch Log Groups Overview

Nine log groups are automatically created across the project's services, including Glue crawler logs, job error/output logs, and session logs, all set to Standard class with a **1-month retention policy**, enabling monitoring and debugging of ETL pipeline execution.



The screenshot shows the AWS CloudWatch Log Groups console. The left sidebar contains navigation links for CloudWatch, Favorites and recents, Ingestion, Dashboards, Monitoring, Alarms (5), AI Operations, GenAI Observability, Application Signals (APM), Infrastructure Monitoring, Logs, Log Management, Log Analytics, and Log Anomalies. The main content area is titled 'Log groups (9)' and includes a search bar, 'Exact match' checkbox, and buttons for 'Actions', 'View in Logs Insights', 'Start tailing', and 'Create log group'. Below is a table of the nine log groups.

☐	Log group	Log class	Anomaly d...	Deletion protection	Bearer token aut...	Data protection	Sensitive data ...	Retention	Metric filters	Contributor Ir
☐	/aws-glue/crawlers	Standard	Configure	Off	Off	-	-	1 month	-	-
☐	/aws-glue/jobs/error	Standard	Configure	Off	Off	-	-	1 month	-	-
☐	/aws-glue/jobs/logs-v2	Standard	Configure	Off	Off	-	-	1 month	-	-
☐	/aws-glue/jobs/output	Standard	Configure	Off	Off	-	-	1 month	-	-
☐	/aws-glue/sessions/error	Standard	Configure	Off	Off	-	-	1 month	-	-
☐	/aws-glue/sessions/output	Standard	Configure	Off	Off	-	-	1 month	-	-
☐	/aws/lambda/RedshiftEventSubscription	Standard	Configure	Off	Off	-	-	1 month	-	-
☐	/aws/lambda/RedshiftOverwatch	Standard	Configure	Off	Off	-	-	1 month	-	-
☐	/aws/lambda/RoleCreationFunction	Standard	Configure	Off	Off	-	-	1 month	-	-